

RETAIL BUSINESS INTELLIGENCE & CUSTOMER INSIGHTS ANALYTICS

Project Report

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Course: Data Analytics

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1. Introduction

Retail businesses generate large volumes of data related to sales, customers, products, stores, and inventory. Analyzing this information helps businesses understand purchasing patterns, monitor performance, and identify growth opportunities. This project develops a Retail Performance and Customer Insights Dashboard to analyze sales performance, customer behaviour, product demand, and inventory status. Interactive dashboards, KPIs, and visualizations are used to present meaningful insights and support effective, data-driven business decisions.

2. Problem Statement

Retail businesses generate large volumes of sales and customer data, making it challenging to identify key trends and performance patterns from raw information. Understanding sales performance, customer behaviour, product demand, and inventory status requires effective data analysis and visualization. This project addresses these challenges by analyzing retail data and presenting meaningful insights through an interactive dashboard. The analysis supports customer engagement, inventory planning, sales improvement, and data-driven business decision-making.

3. Objectives

The main objectives of the project are:

- To clean and prepare retail data using Microsoft Excel.
- To develop a structured relational database using MySQL.
- To analyze sales and customer data using SQL queries.
- To evaluate product, store, staff, and inventory performance.
- To identify high-value customers and purchasing patterns.
- To perform customer segmentation using Python and K-Means clustering.
- To develop interactive dashboards using Microsoft Power BI.
- To generate meaningful business insights and recommendations.
- To support data-driven retail decision-making.

4. Dataset Description

The project uses a structured retail dataset containing multiple business entities, including:

- Customers
- Products
- Orders
- Order Items
- Brands
- Categories
- Staff
- Stocks
- Stores

The dataset contains information related to customer transactions, products, orders, store operations, staff activities, and inventory. The SQL implementation confirms these major tables within the retail database.

The dataset provides a foundation for analyzing sales performance, customer purchasing behaviour, product demand, store performance, and inventory health.

5. Tools and Technologies Used

Tool	Purpose
Microsoft Excel	Data cleaning, validation, and preprocessing
MySQL Workbench	Database creation, data management, and SQL analysis
Python	Data analysis, visualization, and customer segmentation
Microsoft Power BI	Data modeling, KPI development, visualization, and dashboard reporting

The project task sheet defines the same four-tool workflow, including Excel preprocessing, SQL data management, Python analytics and machine learning, and Power BI business intelligence reporting.

6. Project Workflow

The project was completed through a structured end-to-end analytics process:

Data Collection → Excel Data Preparation → SQL Database Development → SQL Analysis → Python Analysis → Customer Segmentation → Power BI Data Modeling → Dashboard Development → Business Insights → Recommendations

The workflow integrates data preparation, database management, analytical querying, machine learning, and business intelligence into a single retail analytics solution.

7. Project Implementation

7.1 Data Preparation Using Microsoft Excel

The retail dataset was initially reviewed and prepared using Microsoft Excel. The purpose of this stage was to improve data quality and ensure consistency before implementing the data in the SQL database.

The preparation process included:

- Reviewing the dataset structure.
- Standardizing column headers.
- Checking data consistency.
- Identifying duplicate records.
- Removing unnecessary duplicates.
- Reviewing missing and inconsistent values.
- Preparing the cleaned dataset for further analysis.

The task sheet specifically identifies column standardization and duplicate removal as part of the Excel preparation phase.

7.2 Database Development Using MySQL

The cleaned data was implemented in MySQL to create a structured relational database. The database was created under the name Retail, with separate tables representing major retail business entities.

The database implementation included:

- Creating the Retail database.
- Creating relational tables.
- Establishing primary and foreign key relationships.
- Managing business records.

- Validating stored data.
- Preparing the database for SQL-based analysis.

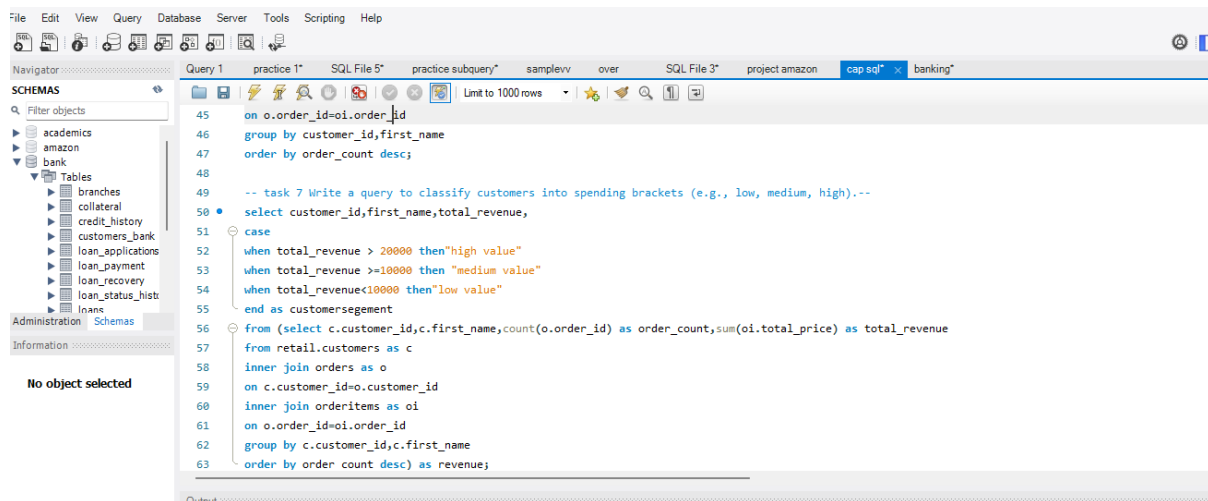
7.3 SQL-Based Business Analysis

SQL was used to retrieve, combine, summarize, and analyze the retail data.

The analysis included:

- Joining Orders, Order Items, and Products to obtain detailed transaction information.
- Calculating total sales for individual stores.
- Identifying the top five products based on quantity sold.
- Calculating customer order counts and total revenue.
- Classifying customers into low-, medium-, and high-value spending groups.
- Analyzing staff-related revenue performance.
- Identifying products with low stock quantities.

SQL functions and techniques such as JOIN, GROUP BY, ORDER BY, aggregate functions, CASE expressions, and subqueries were applied to generate business-oriented results.



8. Customer Analysis Using Python

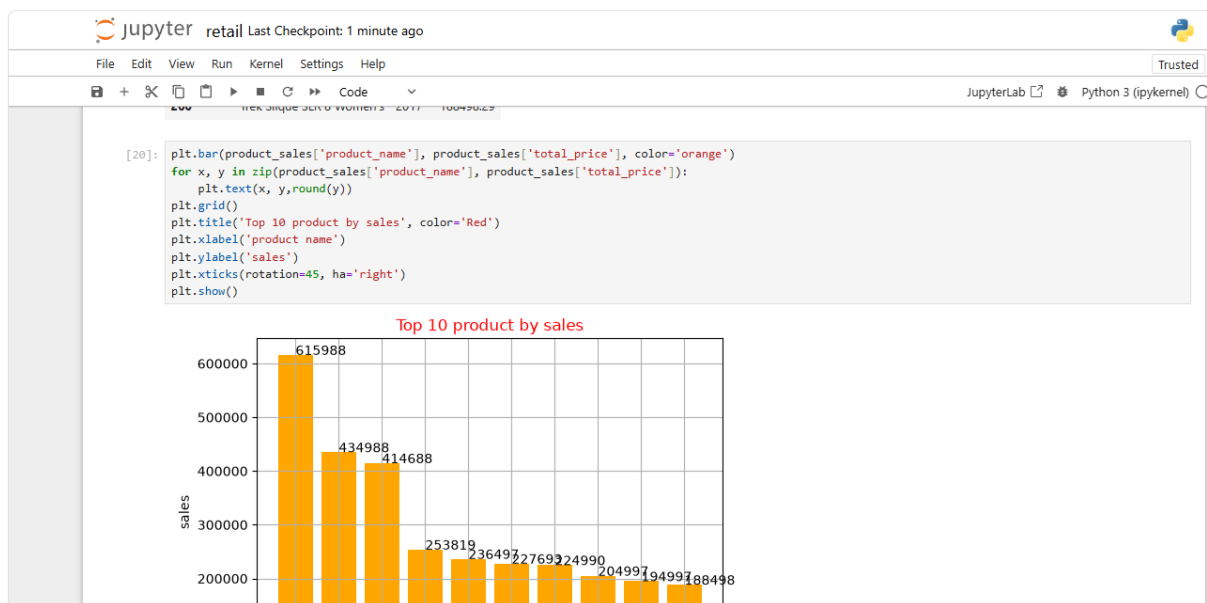
Python was used to perform additional customer and sales analysis. The project task sheet identifies Python-based data analysis, visualization, and customer segmentation as key components of the project.

The Python analysis included:

- Importing the prepared retail data.
- Data preprocessing using Pandas.

- Exploratory Data Analysis.
- Examining customer purchasing behaviour.
- Preparing customer-level analytical features.
- Performing RFM-based customer analysis.
- Applying K-Means clustering for customer segmentation.
- Interpreting customer groups for business use.

Customer segmentation helps distinguish customers according to their purchasing characteristics and provides a basis for targeted marketing and customer retention strategies.



9. Power BI Implementation

The processed retail data was imported into Power BI to create an interactive dashboard. Power Query was used to prepare the data, and relationships were created between the tables.

9.1 Data Preparation

- Imported the processed retail data.
- Checked data types and columns.
- Cleaned and transformed data using Power Query.
- Created relationships between tables.
- Prepared the data for reporting.

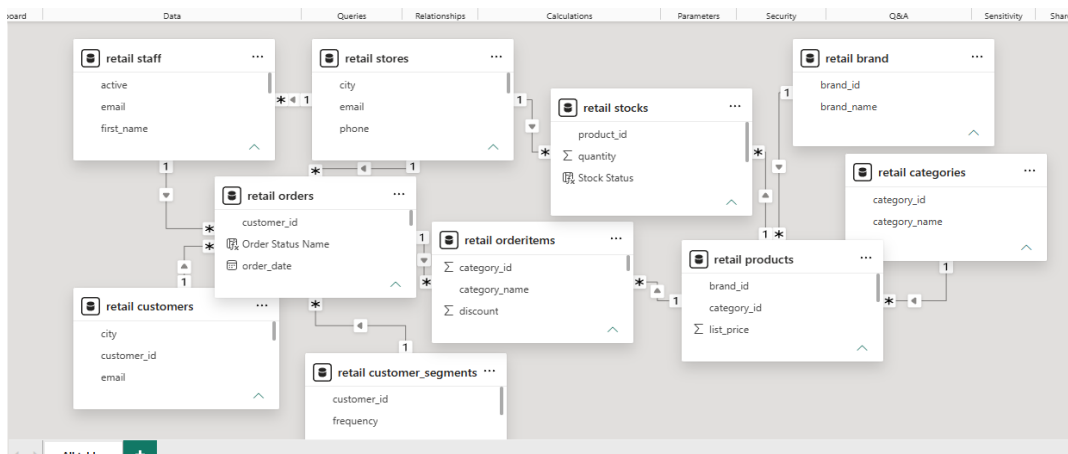
9.2 Dashboard Design

A simple and professional retail dashboard was designed using:

- KPI cards
- Charts and tables
- Slicers and filters
- Navigation buttons
- Consistent formatting

9.3 Data Modeling

- Created relationships between related tables.
- Developed DAX measures for important KPIs.
- Organized the data model for accurate reporting.



9.4 Dashboard Pages

Home / Retail Intelligence Hub

- Displays Total Sales, Orders, Customers, and Average Order Value.
- Provides an overview of overall business performance.
- Includes navigation to other dashboard pages.

Sales Intelligence Hub

- Analyzes sales and order performance.
- Shows store, territory, product, and monthly sales.
- Highlights low-stock products.

Customer Intelligence Hub

- Analyzes customer behaviour and spending.
- Shows top customers and customer segments.
- Compares revenue across customer groups.

9.5 Dashboard Functionality

The dashboard includes:

- KPI Cards, Slicers, Interactive Charts, Cross-filtering, Navigation Buttons, Drill-down, DAX Measures

These features make the dashboard easy to explore and help users understand sales, customers, products, and inventory performance.



10. Business Insights

The completed dashboard provides several important business insights:

- The business records approximately 2K orders and 1K active customers.
- Total sales reach approximately ₹8.58 million.
- The average order value is approximately ₹5.31K.
- Marcelene is identified as the top customer by revenue contribution.
- Regular Customers represent the strongest revenue-generating segment.
- Electra Townie 8i is identified among the strongest-performing products.
- The dashboard identifies 221 inventory items requiring stock monitoring.
- Monthly and annual revenue analysis helps identify changes in sales performance.
- Store and territory analysis provides visibility into regional business performance.

11. Business Recommendations

- Strengthen customer retention by providing personalized offers to high-value customers.

- Target customer segments with differentiated marketing strategies, particularly the Regular Customer segment.
- Increase inventory availability for high-demand products such as Electra Townie 8i.
- Monitor the 221 inventory alerts and establish appropriate reorder levels to reduce stock shortages.
- Improve weaker-performing products and stores through targeted promotions and performance monitoring.
- Use monthly sales trends to plan seasonal campaigns and optimize sales opportunities.
- Monitor Power BI KPIs regularly to support timely and data-driven management decisions.

12. Challenges Faced

The major challenges encountered during the project included:

- Cleaning and standardizing retail datasets.
- Understanding relationships between multiple business tables.
- Designing a structured relational database.
- Writing SQL queries involving multiple tables.
- Performing customer-level analysis.
- Applying customer segmentation techniques using Python.
- Integrating processed data into Power BI.
- Designing dashboards that communicate insights clearly.
- Maintaining consistency across dashboard pages and visualizations.

13. Conclusion

The Retail Sales Analytics Using Excel, SQL, Python and Power BI project successfully demonstrates an end-to-end data analytics workflow. Excel was used for data preparation, MySQL for structured database management and SQL analysis, Python for customer analytics and segmentation, and Power BI for interactive business intelligence reporting. The final solution provides a comprehensive view of sales performance, customer behaviour, product demand, store performance, and inventory health. With ₹8.58M in total sales, 2K orders, 1K active customers, and 221 inventory alerts, the dashboard provides actionable information that can help management improve customer engagement, inventory planning, sales performance, and overall business decision-making.